SIDDHARTH SINGH BHADAURIA

Senior Cloud and Data Engineer | AI Infrastructure Specialist

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Portfolio: siddharthsbhadauria.github.io/portfolio/ • **GitHub:** github.com/siddharthsbhadauria • **LinkedIn:** linkedin.com/in/siddharthsbhadauria

PROFESSIONAL SUMMARY

Senior Cloud and Data Engineer with over 12 years of experience designing and operating cloud infrastructure, data pipelines and AI platforms across Google Cloud Platform (GCP), Microsoft Azure and AWS. Recent work includes building GCP organisation-wide logging, asset inventory and alerting platforms supporting 50M+ daily events, developing core Terraform AWS EKS cluster modules running across 3,600+ deployments and 700+ microservices, and co-engineering enterprise Retrieval-Augmented Generation (RAG) platforms for 75,000+ internal users. Certified Google Professional ML Engineer and Professional Data Engineer using modern AI-assisted engineering tools (GitHub Copilot, Claude Code and Google Antigravity) to build reliable cloud infrastructure and automated workflows.

TECHNICAL CORE COMPETENCIES

- **Cloud Platforms & DevOps:** Google Cloud Platform (GCP), Microsoft Azure, Amazon Web Services (AWS), AWS EKS, GCP Cloud Asset Inventory (CAI), Log Routers, Log Sinks, Log Buckets, GCS Object Lifecycle Management, CMEK, Cloud KMS HSM, BigQuery Linked Datasets, Cloud Run v2, Cloud Functions Gen2, Cloud Scheduler, Artifact Registry, Terraform, Docker, Kubernetes (AKS, GKE), Azure Container Apps, Google Private Service Connect, Azure API Management, GitHub Actions, GitLab CI/CD
- **Generative AI & LLMs:** Retrieval-Augmented Generation (RAG), Vector Databases (Postgres pgvector, Qdrant, ChromaDB), Confident AI DeepEval Framework, Prompt Engineering, Azure OpenAI, GCP Vertex AI, Gemini 1.5 Pro/Flash, Llama 3, Mistral 7B
- **Developer Tools & AI Assistants:** GitHub Copilot, Claude Code, Google Antigravity, VS Code, Git, PyTest, pytest-bdd, Go Test
- **Data Engineering & Streaming:** Data Architecture, Kafka Connect Source Connectors, Enterprise Databus, BigQuery Linked Datasets, Pub/Sub Cloud Storage Subscriptions, Cloud Monitoring Notification Channels, DuckDB, Polars, Great Expectations Data Contracts, BigQuery, Pub/Sub, Splunk Log Ingestion, Cloud Functions, Cloud Scheduler, Azure Data Factory (ADF), dbt Core, Apache Kafka, Apache NiFi, AbInitio, IBM DataStage
- **Databases & Storage:** PostgreSQL, Azure SQL Server, Oracle 12c, Teradata, Sybase, Azure Cosmos DB, AWS S3, DynamoDB, RDS, Google Cloud Storage (GCS), Azure Blob Storage
- **Security & Governance:** CMEK, Cloud KMS HSM, Workload Identity Federation (Keyless), Kerberos-Authenticated Egress Proxy, PII Masking, Fernet Cryptography, GPG Key Management and Rotation, Private DNS Forwarding, Azure Key Vault, Enterprise Data Governance
- **Languages:** Python, Go, SQL, Unix Shell Scripting, PySpark

OPEN-SOURCE ENGINEERING PROJECTS & PORTFOLIO

Cloud-FinOps-RAG | siddharthsbhadauria.github.io/cloud-finops-rag/

- Built an automated pricing pipeline tracking token costs and benchmarks across 15+ LLM provider APIs using GitHub Actions, DuckDB and Polars.
- Enforced data validation contracts with Great Expectations, indexed benchmarks in ChromaDB and published daily summary reports to GitHub Pages.

Auto-DataPulse | github.com/siddharthsbhadauria/auto-datapulse

- Built a telemetry lakehouse collecting system and hardware metrics across 20+ containerised services into DuckDB and Parquet storage.
- Set up schema validation contracts using Great Expectations, data transformation models with dbt Core and containerised deployment with Docker Compose.

Awesome-AI-Discoveries | github.com/siddharthsbhadauria/awesome-ai-discoveries

- Built an automated scraper tracking trending open-source AI, RAG and developer tool repositories on GitHub with an interactive web directory.

Personal Interactive Engineering Portfolio | siddharthsbhadauria.github.io/portfolio/

- Designed and deployed a responsive cloud engineering portfolio showcasing architecture patterns, live project dashboards and open-source contributions.

PROFESSIONAL EXPERIENCE

Senior Cloud Infrastructure & Observability Engineer | Deloitte (Client: JPMorgan Chase & Co.) — **London, UK | June 2026 - Present**

JPMorgan Chase (GCP Cloud Infrastructure, Enterprise Observability & Security)

- Built organisation-wide log routing pipelines in Terraform across audit, network flow and platform telemetry, fanning out organisation-level sinks into 16 Pub/Sub topics for streaming to SIEM and Kafka Databus.

- Cut long-term log storage costs by ~94% using GCS lifecycle policies that move data to Archive tier after 90 days and enforce deletion at 540 days.
- Standardised Cloud KMS HSM encryption keys (CMEK) with 30-day rotation across all storage and streaming services (GCS, Cloud Logging, BigQuery and Pub/Sub).
- Designed an active/standby two-region asset inventory export pipeline running 12 scheduled batch jobs daily on Cloud Run v2 and Cloud Functions Gen2, maintaining a 2-hour data freshness target.
- Migrated the export service to Cloud Run v2 with zero downtime using container digest validation, Direct VPC Egress and keyless Workload Identity Federation.
- Built an alert transport pipeline routing Cloud Monitoring alerts to Kafka and ServiceNow via Pub/Sub and time-partitioned GCS storage, keeping delivery delay under 60 seconds.
- Shipped 190+ production pull requests across 3 infrastructure repositories in 7 months, using environment-agnostic Terraform blueprints, automated BDD test suites and pipeline monitoring.

Parental Leave | Career Break **London, UK | Oct 2025 - May 2026**

- Took planned parental leave following the completion of the AWS EKS cluster deployment project.

Senior Consultant & DevOps Engineer | Deloitte (Client: JPMorgan Chase & Co.) **London, UK | Aug 2024 - Sept 2025**

JPMorgan Chase (Terraform AWS EKS & Microservices Platform)

- Maintained and improved core Terraform AWS EKS cluster modules alongside microservices blueprints connecting EKS to S3, DynamoDB, RDS, MSK, SQS and SNS, reducing provisioning time by 65%.
- Scaled the EKS module to 3,600+ deployments supporting 700+ microservices across 20+ business lines.
- Transitioned the cluster module into a multi-tenant model to streamline resource allocation and tenant isolation.
- Managed Kubernetes version upgrades from v1.19 through v1.34 with documented migration guides, supporting 50+ zero-downtime production releases.
- Built automated validation test suites using Python and Go to test module changes before deployment.

Senior Consultant | Deloitte (Internal: PairD GenAI Platform) **London, UK | May 2024 - Aug 2024**

Deloitte PairD (Enterprise Generative AI Platform)

- Co-developed Deloitte's internal AI platform for 75,000+ employees, handling over 100,000 daily queries across OpenAI and Google Gemini models.
- Set up secure networking using Google Private Service Connect (PSC) and custom DNS forwarding for full network isolation, keeping model traffic strictly on internal networks.
- Built RAG pipelines with Postgres pgvector and Qdrant, tuning prompt templates to improve retrieval accuracy.
- Converted Azure cloud workloads into modular Infrastructure as Code using Terraform and ARM templates.
- Configured Docker containerised services deployed to Azure Container Apps and Azure Kubernetes Service (AKS).
- Improved development cycle times by adopting AI-assisted workflows with GitHub Copilot, Claude Code and Google Antigravity.
- Set up CI/CD deployment pipelines using GitHub Actions and Azure Container Registry, exposing backend services through Azure API Management.

Senior Consultant | Deloitte (Internal: GenAI LLM Evaluation) **London, UK | Apr 2024 - May 2024**

GenAI LLM Evaluation Framework

- Built an automated evaluation pipeline using Confident AI DeepEval to measure model quality across answer relevancy, contextual accuracy, completeness and latency.
- Benchmarked Mistral 7B, Llama 3 and Gemini 1.0 Pro on GCP Vertex AI across standardized enterprise test prompts.
- Built Python scripts to collect and visualize benchmarking metrics, reducing evaluation turnaround from 2 weeks to under 4 hours.

Architect & Senior Developer | Deloitte (Internal: DataSAFE Accelerator) **London, UK | Dec 2023 - Mar 2024**

DataSAFE & M&A Data Transfer Accelerator

- Built data processing pipelines for 10TB+ of GPG-encrypted datasets, including schema validation, integrity checks and automated decryption.
- Developed Python scripts to identify and mask Personally Identifiable Information (PII) across 1,000+ database columns using Fernet cryptography.
- Configured staging in Azure Blob Storage, schema management in Azure Cosmos DB and key management in Azure Key Vault.
- Built Azure Functions for event-driven, controlled unmasking of datasets for authorized consumers.

Azure Data Engineer | Deloitte (Client: UBS AG) **London, UK | Jul 2023 - Aug 2023**

- Built migration and archival pipelines moving legacy Credit Suisse application data into UBS Azure infrastructure across 1,000+ systems.
- Built Azure Functions and Azure Data Factory (ADF) pipelines, increasing ingestion throughput 3x from PostgreSQL, Sybase and Oracle databases.
- Provisioned cloud infrastructure using Terraform and set up CI/CD workflows using GitLab CI/CD and Azure DevOps.
- Designed event-driven file validation, GPG encryption and archival workflows meeting financial compliance and legal hold requirements.
- Built tracking dashboards using GitLab, Power BI and Alteryx to monitor migration status and pipeline health.

Senior Cloud Data Engineer | Deloitte (Client: Virgin Media O2) **London, UK | Jul 2022 - Apr 2023**

- Built ingestion pipelines loading HR records for 40,000+ employees from Oracle HCM, Workday and SuccessFactors into GCP BigQuery.
- Enforced data security with GPG encryption, role-based access controls and custom Python scripts for automated PII masking.
- Wrote shell scripts for automated GPG key generation and rotation schedules, reducing maintenance overhead by 80%.
- Developed AbInitio graphs and integrated dbt transformation workflows to standardize BigQuery data structures for analytics.

Data Engineer | Deloitte (Client: Motor Insurers' Bureau) **London, UK | Mar 2021 - May 2021**

- Built a data quality scoring platform on Azure for 3 core insurance databases (MID, CUE and MIAFTR) covering 35M+ records.
- Automated environment provisioning via Terraform, deploying Azure Data Factory, Azure SQL Server, Storage Accounts and Informatica Secure Agents.
- Configured SFTP integration pipelines to ingest feeds from Experian, CAP HPI and Solera into Azure SQL Server.
- Implemented Informatica Cloud (IICS) data quality profiling and validation rules, generating automated scorecards for business stakeholders.

Senior Cloud & Data Engineer | HSBC **London, UK / Pune, India | Jan 2015 - Nov 2021**

- CESOP Reporting: Built an AWS to GCP migration proof-of-concept for financial payment reporting, writing Terraform scripts for GCP DataProc and Cloud Functions.
- Real-Time Streaming: Built streaming pipelines processing 10M+ daily messages from Apache Kafka to BigQuery via GCS using Apache NiFi.
- Change Data Capture: Set up IBM CDC (DB2) and Attunity Replicate (Oracle) streaming via Kafka Connect and AWS Lambda for cross-cloud transfers between AWS and GCP.
- Legacy Data Migration: Led data transformation pipelines using IBM DataStage, Hadoop, Hive, Sqoop and Oracle for Common Data Repository (CDR) migration.

CERTIFICATIONS (CREDLY VERIFIED)

- **Google Cloud Certified: Professional Machine Learning Engineer** (Issued Dec 2025 - Dec 2027)
- **Google Cloud Certified: Professional Data Engineer** (Issued Dec 2025 - Dec 2027)
- **Google Cloud Certified: Generative AI Leader** (Issued Jan 2026 - Jan 2029)
- **Google Cloud Certified: Associate Data Practitioner** (Issued Jan 2026 - Jan 2029)
- **Google Cloud Certified: Cloud Digital Leader** (Issued Feb 2024 - Feb 2027)
- **AWS Certified AI Practitioner** (Issued Sep 2024 - Sep 2027)
- **HashiCorp Certified: Terraform Associate (003)** (Issued Sep 2024 - Sep 2026)
- **AWS Certified Cloud Practitioner** (Issued Jul 2022 - Feb 2028)
- **Microsoft Certified: Azure Fundamentals (AZ-900)** (2024)
- **Microsoft Certified: Data Fundamentals (DP-900)** (2024)
- **Oracle Cloud Infrastructure: Certified Foundations Associate** (2024)

EDUCATION & LANGUAGES

Bachelor of Engineering in Electronics and Communication | RGPV University, India (2010 - 2014)
Languages: English (Fluent), Hindi (Native)